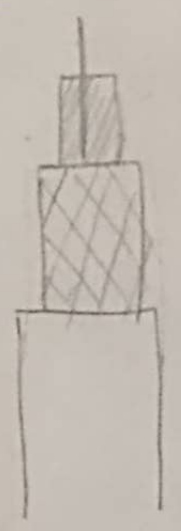
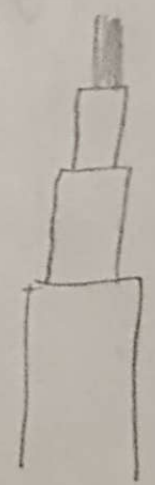
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2. STUDY OF NETWORK CABLES

Aim:

To study of different types of network cables.

Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Application/ Use	Image
UTP	Category 3	10 Mbps	Advantages: • Cheaper in cost • Easy to install as they have a smaller overall diameter Disadvantages: • More prone to EMI and noise	10 Base-T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, Gigabit Ethernet	
	Category 5	Up to 100 Mbps			
	Category 5e	1 Gbps			
STP	Category 6,6a	10 Gbps	Advantages • Shielded, Faster than UTP, Less susceptible to noise and inference Disadvantages: • Expensive, Greater installation effort	1 GB Ethernet, 10 G Ethernet (55m) Widely used in data centres GB Ethernet, 10 G Ethernet (100 m)	
SSTP	Category 7	10 Gbps			
Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	Advantages: • High bandwidth, Immune to inference Low loss bandwidth Disadvantages Limited distance, cost, size is bulky.	Speed of signal is 500m Television network High speed internet connections	
Fibre optics cable	Single mode multi mode	100 Gbps	Advantages: High speed, High bandwidth, High security, long distance Disadvantages: Expensive, Requires skilled installers	Maximum distance of fibre optics cable is around 100 meters	

Make your own ethernet cross over cable / straight cable:

step 1: To start construction of the device, begin by threading shields onto the cable.

step 2: Next, strip approximately 1.5cm of cable shielding from both ends. The crimping tool on has a round area to complete task.

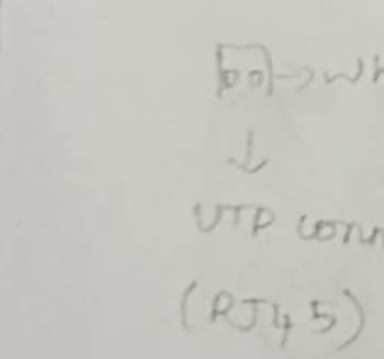
step 3: After, you will need to untangle the wires. There should be four 'twisted pairs.' Refrencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.

step 4: Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others.

step 5: Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding.

step 6: After the wires are securely sitting inside the pl insert it into the crimping tool and push down.

step 7: Lastly, repeat for the other end using diagram B usi diagram A.



Color order

pin 1: White-
pin 2: Orange
pin 3: White-
pin 4: Blue
pin 5: White-
pin 6: green
pin 7: White
pin 8: Brown

Result:

Thus, all kind of cables are studied.